

DL-Project-Label-Flipping

Label Flipping Attacks in Federated Learning: URLF vs FBPLF on NSL-KDD

Deep Learning Course Project

1. Introduction

Federated Learning (FL) allows multiple clients to collaboratively train a shared model without exchanging raw data. This decentralization, however, opens the door to data poisoning attacks: a malicious client can corrupt its local training labels before participating in a training round, biasing the global model.

This project implements and compares two label-flipping strategies in a simulated FL system:

Uniform Random Label Flipping (URLF): a malicious client randomly replaces a fraction of its labels with a uniformly drawn different class. This is a noisy, untargeted attack.

Feature-Based Probabilistic Label Flipping (FBPLF): a malicious client computes the feature centroid of each class on its local data, then flips a label probabilistically toward the class whose centroid is closest in feature space (inverse-distance weighting). This produces flips that are semantically plausible -- e.g. a Denial-of-Service sample is more likely to be relabeled as a Probe (reconnaissance) sample than as Normal traffic, since DoS and Probe share similar high-volume traffic characteristics.

The goal is to study whether FBPLF produces a different, more 'targeted' and potentially stealthier degradation pattern than the random baseline (URLF), and to quantify both attack impact (accuracy, attack success rate, targeted misclassification) and detectability (PoD via cosine similarity of client weight updates).

2. Dataset

NSL-KDD is a refined version of the KDD Cup 1999 network intrusion dataset (Tavallae et al., 2009), with duplicate records removed. Each record has 41 features describing a network connection (3 categorical: protocol type, service, flag; remainder numeric), label-encoded and standardized.

This project restricts evaluation to the three dominant classes -- **Normal**, **DoS**, and **Probe** -- which together make up 99.3% of the training data. The R2L (995 samples) and U2R (52 samples) classes are excluded: they are too sparse for FBPLF to compute a reliable local feature centroid in a federated, non-IID partition.

Split	Normal	DoS	Probe	Total
Training	67,343 (53.5%)	45,927 (36.5%)	11,656 (9.3%)	125,973*
Test	9,711 (43.1%)	7,460 (33.1%)	2,421 (10.7%)	22,544*

Table 1. Class distribution (3-class subset). *Total includes the small R2L/U2R remainder not shown above; percentages are of the full set.

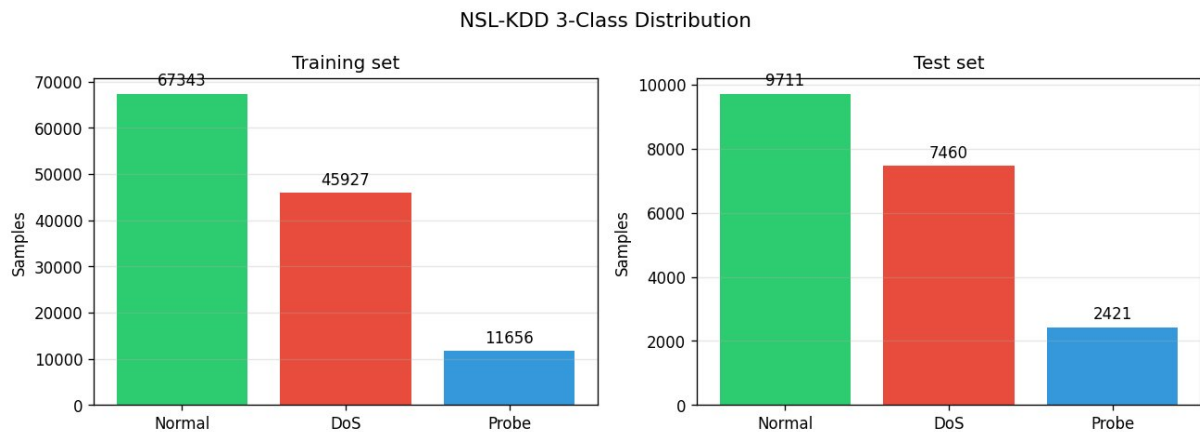


Figure 1. Class distribution, training and test sets.

3. Methodology

3.1 Federated Learning Setup

Parameter	Value
Model	5-layer deep MLP, ~450K parameters: 41 -> 512 -> 512 -> 256 -> 128 -> 3, BatchNorm+ReLU per hidden layer
Aggregation	FedAvg (McMahan et al., 2017)
Clients	50
Rounds	100
Local epochs	1 per round
Optimizer	Adam (lr = 0.001)
Loss	CrossEntropyLoss, class-weighted (inverse frequency)
Client partition	Non-IID via Dirichlet(alpha=0.5)
Malicious clients (N _{mal})	0, 10, 20, 30, 40, 50
Flip fraction (p)	0.5, 0.8, 1.0
Seeds	11, 22, 33

Table 2. Experimental configuration.

3.2 Attack Implementations

URLF: for each malicious client, a fraction p of local samples are selected at random; each selected label is replaced with a uniformly random different class.

FBPLF: for each malicious client, class centroids (mean feature vector per class) are computed on the client's local data. A fraction p of samples are selected; for each, the inverse Euclidean distance to every class centroid is computed and normalized into a probability distribution, and a new label is sampled from it. Closer centroids receive higher probability -- so labels tend to flip toward the most feature-similar class.

Detection (PoD): at the final training round, each client's weight update (local weights minus global weights, excluding BatchNorm running statistics) is flattened into a vector. Cosine similarity to the mean benign update is computed for every client. A detection threshold is set at the 5th percentile of benign similarities (target FPR = 0.05); any client below this threshold is flagged. PoD is the fraction of malicious clients flagged.

4. Results

4.1 Baseline (N_{mal}=10, p=0.5)

Scenario	Accuracy	F1 (macro)	F1 (DoS)	F1 (Probe)	AUROC	ASR	DoS→Probe
Clean	0.8735	0.8365	0.8723	0.7326	0.9301	0.1762	0.0307
URLF	0.8630	0.8248	0.8617	0.7188	0.9350	0.2027	0.0295
FBPLF	0.8630	0.8267	0.8585	0.7265	0.9565	0.1958	0.0315

Table 3. Baseline results (single seed=11). At this moderate poisoning level (20% malicious clients, 50% flip fraction), both attacks have only a small effect on global metrics -- FedAvg's averaging over 40 benign clients absorbs the impact of 10 poisoned ones. Differences become pronounced only at higher poisoning intensity (Section 4.3).

NSL-KDD 3-Class Confusion Matrices ($N_{\text{mal}}=10$, $p=0.5$)

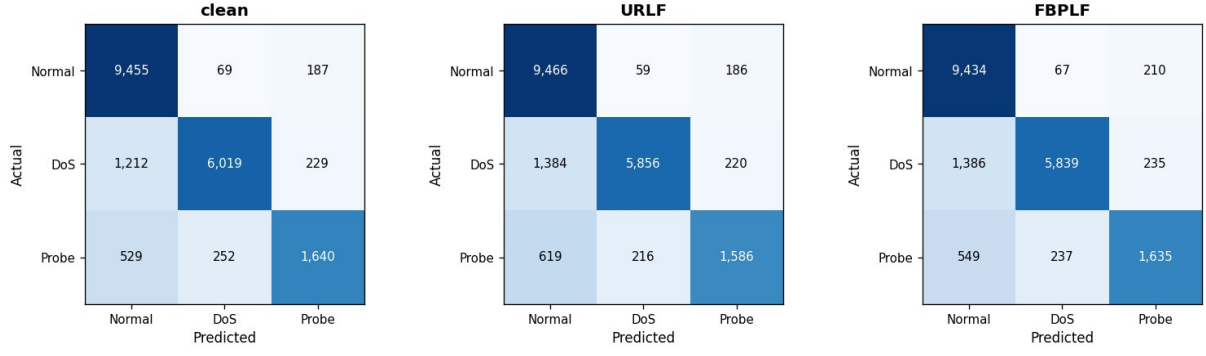


Figure 2. Confusion matrices for clean, URLF, and FBPLF ($N_{\text{mal}}=10$, $p=0.5$).

4.2 Multi-Seed Stability (seeds 11, 22, 33)

Scenario	Accuracy	F1 (macro)	F1 (DoS)	F1 (Probe)	AUROC	ASR
Clean	0.8710±0.0049	0.8299±0.0076	0.8757±0.0050	0.7133±0.0182	0.9360±0.0067	0.1865±0.0105
URLF	0.8703±0.0136	0.8310±0.0216	0.8742±0.0118	0.7197±0.0472	0.9469±0.0112	0.1893±0.0254
FBPLF	0.8721±0.0130	0.8354±0.0169	0.8716±0.0132	0.7320±0.0298	0.9575±0.0014	0.1789±0.0238

Table 4. Multi-seed summary (mean±std, $N_{\text{mal}}=10$, $p=0.5$). Results are stable across independent Dirichlet partitions and random seeds.

4.3 Attack Sweep: Accuracy and ASR vs Poisoned Clients

The sweep covers all combinations of N_{mal} in $\{0,10,20,30,40,50\}$ and p in $\{0.5, 0.8, 1.0\}$. Figure 3 shows global accuracy and Figure 4 shows Attack Success Rate (ASR, fraction of true attack samples -- DoS or Probe -- predicted as Normal).

NSL-KDD 3-Class: Accuracy vs Poisoned Clients

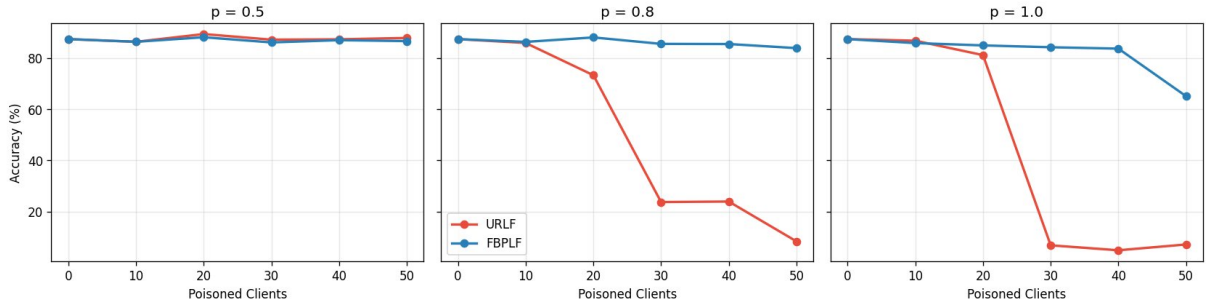


Figure 3. Accuracy vs. number of poisoned clients, for $p=0.5, 0.8, 1.0$. At $p=0.5$ both attacks track the clean baseline closely. At $p \geq 0.8$ and $N_{\text{mal}} \geq 30$, URLF collapses sharply (down to 6.8% accuracy at $p=1.0$, $N_{\text{mal}}=30$) while FBPLF remains stable (84.2% under the same conditions).

NSL-KDD 3-Class: Attack Success Rate vs Poisoned Clients

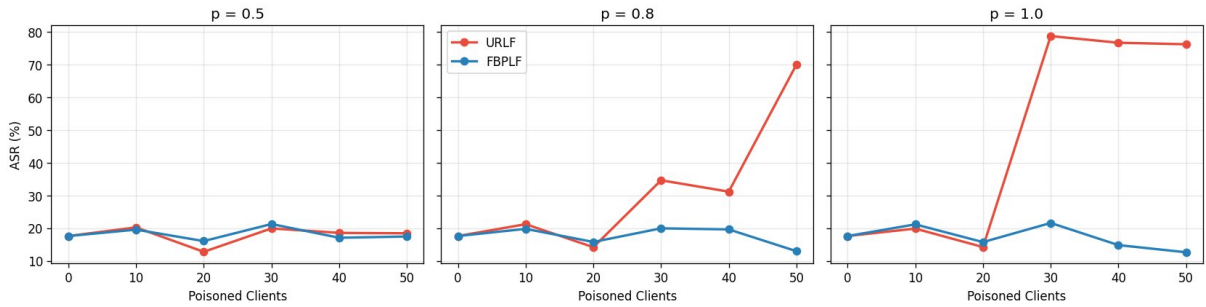


Figure 4. Attack Success Rate vs. number of poisoned clients. At extreme poisoning ($p=0.8-1.0$, $N_{\text{mal}} \geq 30$), URLF's ASR rises to 70-79%, while FBPLF's ASR stays controlled at 13-22%, despite both attacks targeting the same fraction of labels.

Setting	URLF Acc	FBPLF Acc	URLF ASR	FBPLF ASR
p=0.8, N_mal=30	0.2370	0.8549	0.3467	0.1998
p=0.8, N_mal=50	0.0833	0.8383	0.7004	0.1299
p=1.0, N_mal=30	0.0678	0.8417	0.7876	0.2162
p=1.0, N_mal=40	0.0485	0.8363	0.7672	0.1485
p=1.0, N_mal=50	0.0711	0.6504	0.7625	0.1266

Table 5. URLF vs FBPLF at high poisoning intensity, extracted from the sweep. URLF causes catastrophic, easily-noticeable degradation; FBPLF preserves global accuracy while still achieving meaningful ASR.

4.4 Targeted Misclassification (DoS -> Probe)

This is the central result of the project: does FBPLF actually concentrate misclassifications toward the feature-similar class (Probe), rather than scattering them randomly like URLF? Figure 5 shows the prediction distribution for true-DoS test samples under each scenario; Figure 6 tracks the DoS->Probe rate across the full sweep.

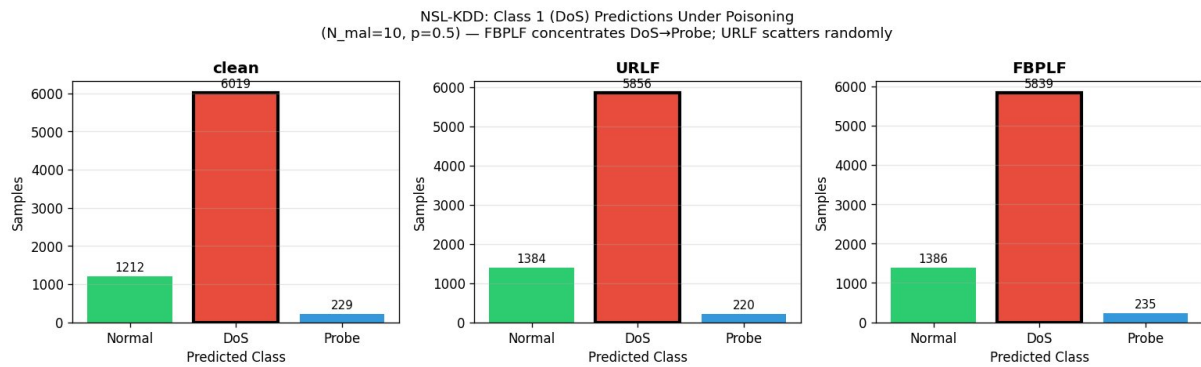


Figure 5. Prediction distribution for true DoS samples (N_mal=10, p=0.5). At this moderate setting the three scenarios are visually similar -- the effect becomes visible only at higher poisoning intensity (Figure 6).

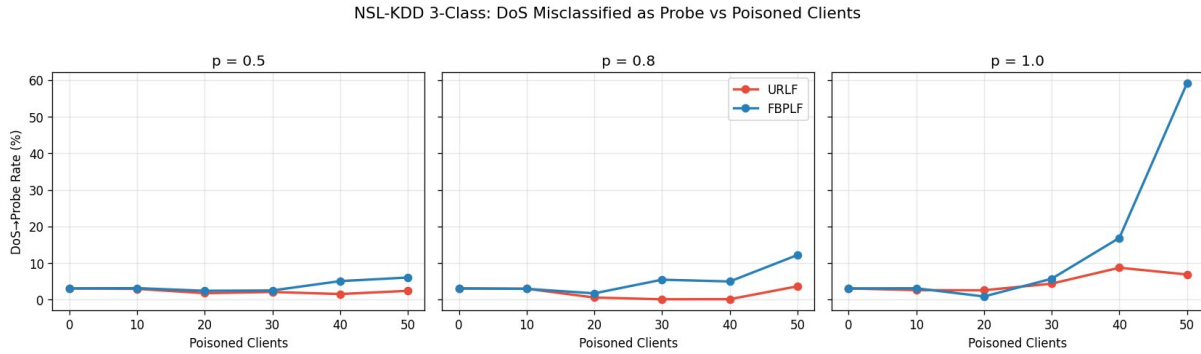


Figure 6. DoS misclassified as Probe (%) vs. number of poisoned clients. At p=1.0, N_mal=50, FBPLF drives the DoS->Probe rate to 59.2%, an 8.6x increase over URLF's 6.9% under the same conditions. This is the tabular analogue of a targeted cat->dog confusion in image classification: FBPLF does not just degrade accuracy, it steers errors toward a specific, feature-similar class.

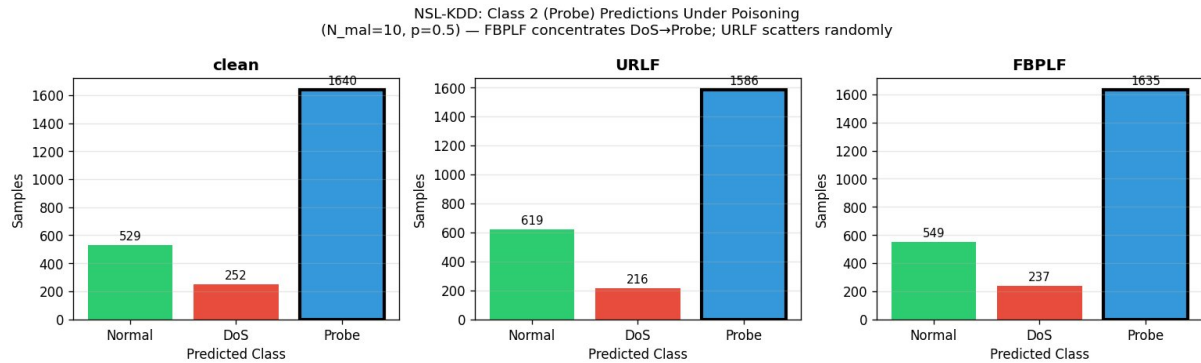


Figure 7. Prediction distribution for true Probe samples (N_mal=10, p=0.5), shown for completeness -- Probe is the smallest class (9.3% of training data).

4.5 Detection Analysis (Probability of Detection)

PoD is measured via cosine similarity between each client's final-round weight update and the mean benign update, with a detection threshold set at the 5th percentile of benign similarities (target FPR=0.05).

Scenario	N _{mal}	Poisoned (%)	PoD	FPR	Threshold
URLF	10	20%	1.0	0.0500	0.1242
FBPLF	10	20%	1.0	0.0500	0.1058
URLF	20	40%	1.0	0.0667	0.0718
FBPLF	20	40%	1.0	0.0667	0.1316
URLF	30	60%	1.0	0.0500	0.4613
FBPLF	30	60%	1.0	0.0500	0.4407
URLF	40	80%	1.0	0.1000	0.4829
FBPLF	40	80%	1.0	0.1000	0.4183

Table 6. PoD results ($p=0.5$). Both attacks are fully detected ($PoD=1.0$) across all configurations -- in this ~450K-parameter MLP, any label perturbation produces a gradient update that deviates noticeably from the benign distribution. (N_{mal}=50, i.e. 100% malicious clients, has no benign clients to form a reference and is omitted.)

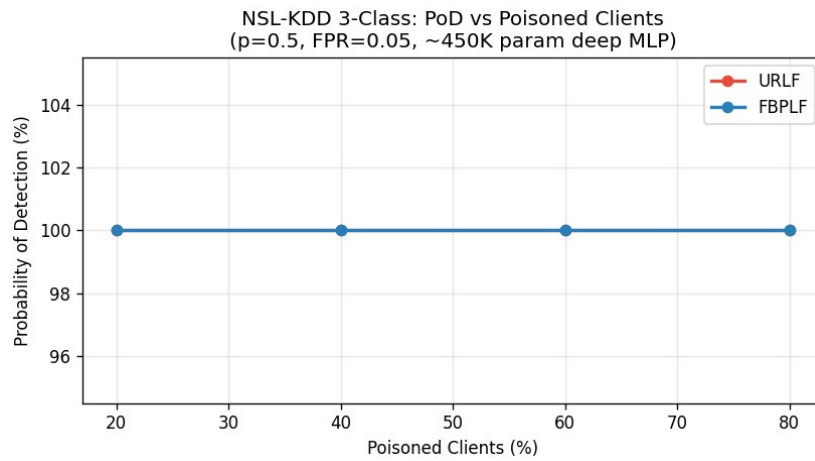


Figure 8. PoD vs. poisoned clients ($p=0.5$).

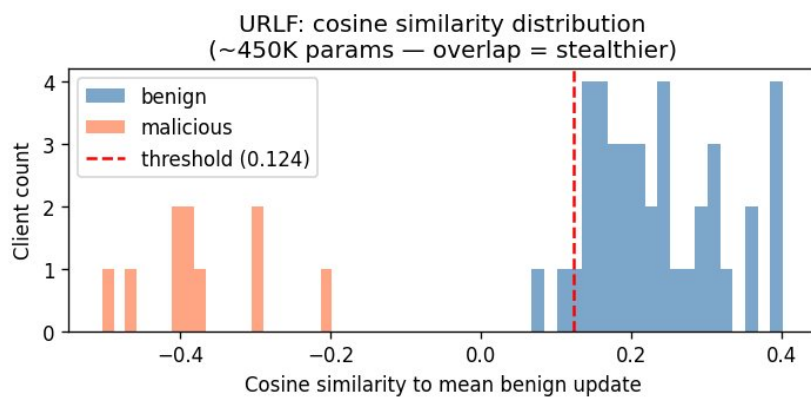


Figure 9. Cosine similarity distribution, URLF (N_{mal}=10, $p=0.5$).

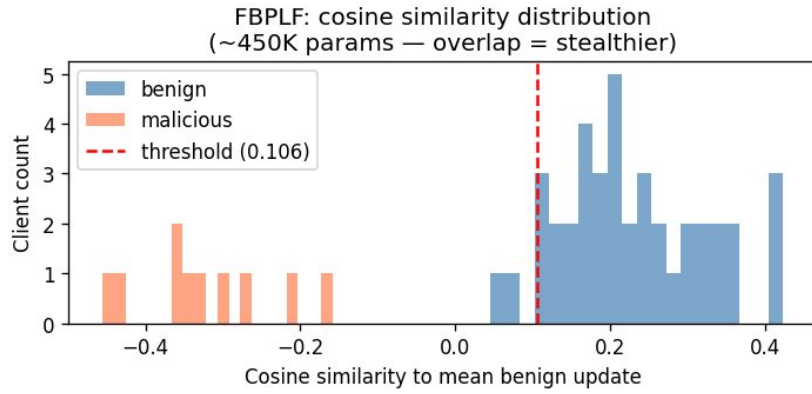


Figure 10. Cosine similarity distribution, FBPLF ($N_{\text{mal}}=10$, $p=0.5$).

Although $\text{PoD}=1.0$ for both, the detection *threshold* is consistently lower for FBPLF (e.g. 0.106 vs 0.124 at $N_{\text{mal}}=10$), indicating FBPLF's update vectors sit directionally closer to the benign mean -- a weaker but real stealth signal. The stronger evidence for FBPLF's relative stealth is the accuracy-preservation result in Section 4.3: a server monitoring only global accuracy would notice URLF immediately (a collapse to $<10\%$) but would see no obvious signal from FBPLF (accuracy stays in the 65-87% range).

5. Conclusion

Targeted vs random degradation. FBPLF concentrates DoS misclassifications toward the feature-similar Probe class (up to 59.2% at extreme poisoning), while URLF scatters errors with no semantic structure (6.9% in the same setting). This confirms the core hypothesis on tabular network traffic data.

Accuracy preservation. At high poisoning intensity ($N_{\text{mal}} \geq 30$, $p \geq 0.8$), URLF collapses global accuracy to single digits, while FBPLF retains 65-88% accuracy -- a direct parallel to URLF/FBPLF behaviour reported for CIFAR-10 under full-scale poisoning.

ASR divergence. URLF's Attack Success Rate climbs to 70-79% under extreme poisoning; FBPLF stays controlled at 13-22%, despite identical flip fractions.

Detection. Cosine-similarity PoD detects both attacks fully ($\text{PoD}=1.0$) in this $\sim 450\text{K}$ -parameter MLP setting. FBPLF nonetheless shows a consistently lower detection threshold, and -- more importantly -- avoids the obvious accuracy-collapse signal that URLF produces.

Overall. FBPLF behaves as a precision attack: it achieves meaningful, targeted misclassification while remaining far less disruptive (and far less obvious) to global model performance than a random baseline.